

Testing for *ESR1* Mutations to Guide Therapy for Hormone Receptor–Positive, Human Epidermal Growth Factor Receptor 2–Negative Metastatic Breast Cancer: ASCO Guideline Rapid Recommendation Update

Harold J. Burstein, MD, PhD¹; Angela DeMichele, MD²; Mark R. Somerfield, PhD³; and N. Lynn Henry, MD, PhD⁴; for the Biomarker Testing and Endocrine and Targeted Therapy in Metastatic Breast Cancer Expert Panels

ASCO Rapid Recommendations Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the ASCO Guideline Methodology Manual. The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options. See the Appendix for disclaimers and other important information (Appendix 1 and Appendix 2, online only).

BACKGROUND

Two previous ASCO guidelines discussed the role of testing in activating gain-of-function mutations in the estrogen receptor gene ligand-binding domain (*ESR1* mutation) to guide therapy for hormone receptor–positive, human epidermal growth factor receptor 2 (HER2)–negative metastatic breast cancer (MBC)^{1,2} and concluded that data were insufficient to recommend routine testing.³ The EMERALD trial, an international, open-label, randomized phase III trial that evaluated the efficacy and safety of the oral selective estrogen receptor degrader, elacestrant, versus endocrine monotherapy (standard-of-care [SOC]) in patients with previously treated estrogen receptor (ER)–positive, HER2-negative advanced breast cancer, including patients with *ESR1*-mutated tumors, provides a strong signal for updating previous ASCO MBC guidelines.

METHODS

A targeted electronic literature search was conducted to identify any additional phase III randomized controlled trials in this patient population. No additional randomized controlled trials were identified. The original guideline Expert Panels reconvened to review evidence from EMERALD and to review and approve the revised recommendations.

EVIDENCE REVIEW

EMERALD randomly assigned postmenopausal women or men with advanced breast cancer to either elacestrant 345 mg orally once daily (n = 239) or physician's choice SOC endocrine therapy (ET) with fulvestrant, anastrozole, letrozole, or exemestane monotherapy (n = 238); the

SOC control arm permitted reintroduction of previous therapy and did not permit combination therapy with targeted agents such as alpelisib or everolimus. Eligible patients had cancer progression after first- or second-line ET and had received a CDK4/6 inhibitor in combination with ET as part of their previous therapy. The coprimary end points were progression-free survival (PFS) in all patients (N = 477) and in patients with detectable *ESR1* mutations in circulating tumor DNA (ctDNA; n = 228). Compared with SOC, there was improved PFS with elacestrant in both the overall study population (hazard ratio [HR], 0.70; 95% CI, 0.55 to 0.88; *P* = .002) and patients with *ESR1* mutations in ctDNA (HR, 0.55; 95% CI, 0.39 to 0.77; *P* = .0005). In cases with *ESR1* mutations, the median PFS was 3.8 months for elacestrant and 1.9 months for SOC, and 6-month PFS rates were 41% and 19% for elacestrant and SOC, respectively. Among patients without detectable *ESR1* mutations, there was no significant improvement in PFS with elacestrant compared with SOC ET (HR, 0.86; 95% CI, 0.63 to 1.19; *P* = .308). Objective response rates were 4% in each arm in the overall population and were not statistically significantly different between elacestrant and SOC ET in those with detectable *ESR1* mutations (7.1% v 4.7%; *P* = .455). Elacestrant therapy was associated with more toxicity than SOC ET including nausea and vomiting. Appraisal of the trial report using the GRADE⁴ instrument was performed as per ASCO's methodology and found a high certainty of the evidence.

UPDATED RECOMMENDATIONS

To aid in treatment selection, the Expert Panel recommends routine testing for emergence of *ESR1* mutations at recurrence or progression on ET (given with or

ASSOCIATED CONTENT

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Appendix

Author affiliations and support information (if applicable) appear at the end of this article.

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without CDK4/6 inhibitor) in patients with ER-positive, HER2-negative MBC. Testing with a Clinical Laboratory Improvement Amendments–certified assay should be performed on blood or tissue obtained *at the time of progression*, as *ESR1* mutations develop in response to selection pressure during treatment and are typically undetectable in the primary tumor.⁵ Blood-based ctDNA is preferred owing to greater sensitivity.⁶ If not performed earlier, testing for *PIK3CA* mutations should also be performed to guide further therapy. Patients whose tumor or ctDNA tests remain *ESR1* wild-type may warrant retesting at subsequent progression(s) to determine if an *ESR1* mutation has arisen (Type: Evidence-based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Patients previously treated with ET and a CDK4/6 inhibitor for advanced breast cancer have several therapeutic options if

choosing to continue endocrine-based approaches. For patients with prior CDK4/6 inhibitor treatment and *ESR1* wild-type tumors, appropriate subsequent ET options include fulvestrant, aromatase inhibitor, or tamoxifen monotherapy, or ET in combination with targeted agents such as alpelisib (for *PIK3CA*-mutated tumors), or everolimus. For patients with prior CDK4/6 inhibitor treatment and a detectable *ESR1* mutation, options include elacestrant, or other ET either alone or in combination with targeted agents such as alpelisib (for *PIK3CA*-mutated tumors) or everolimus. While elacestrant has comparable or greater activity than SOC ET monotherapy, there are at present no data on the safety or clinical efficacy to support its use in combination with targeted agents.

For all guideline recommendations, a complete summary table is available at www.asco.org/breast-cancer-guidelines.

AFFILIATIONS

¹Dana Farber Cancer Institute, Boston, MA

²University of Pennsylvania, Philadelphia, PA

³American Society of Clinical Oncology, Alexandria, VA

⁴University of Michigan, Ann Arbor, MI

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Rd, Ste 800, Alexandria, VA 22314; e-mail: guidelines@asco.org

EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at www.asco.org/breast-cancer-guidelines.

EQUAL CONTRIBUTION

H.J.B., A.D., and N.L.H. were Expert Panel coauthors. H.J.B. and A.D. contributed equally to this work as first authors.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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AUTHOR CONTRIBUTIONS

Conception and design: All authors

Administrative support: Mark R. Somerfield

Collection and assembly of data: All authors

Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

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Harold J. Burstein

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Angela DeMichele

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N. Lynn Henry

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